

Frank Shan

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HIGHLIGHTS

- **Full-stack** developer specialized in translating **business requirements into production-ready** systems. Shipped a complete product in under 3 months. Expert in **building MVP** rapidly using coding agents.

EDUCATION

University of Waterloo | *CGPA: 92.3/100* Sep 2024 - Apr 2029
Bachelor of Computer Science, Honours Computer Science, AI Specialization

EXPERIENCE

Software Developer Intern | *Richmond, BC* Sep 2025 - Apr 2026
Ziyutec Inc.

- **Owned** the end-to-end planning and development of a [global B2B marketplace](#) for a startup initiative, **leading** system design, implementation, and deployment (AWS) across the **entire SDLC**.
- Architected a Role-Based Access Control framework (NestJS, PostgreSQL) to support extensible user roles; balanced database **normalization** against **query performance** via denormalization and data **integrity** checks.
- Engineered a **caching** layer to decouple client-side state from **atomic** database transactions, reducing **user friction** while ensuring transactional integrity.
- Built an automation agent for engineering workflow, reducing operational workload by **20%** across a **50-person** engineering team.
- Built a **RAG agent**, indexing **500+** **enterprise documents** of different format; improved **contextual relevance by 16%** (DeepEval) via **hybrid search and reranking** on synthetically generated evaluation set.
- Maintained and enhanced a **multi-tenant** ERP platform comprising **60K+** **lines of legacy code**, supporting **1,000+** **active projects**. Resolved a **long-standing** session management defect by implementing **JWT refresh token rotation**, eliminating form data loss.

Research Assistant | *Waterloo, ON* Feb 2026 - Present
University of Waterloo

- Designed comparative experiments for a probabilistic programming language PAL that compiles structural rules into tensor operations. Built a **hybrid neuro-symbolic** framework for training Named Entity Recognition models to enforce strict linguistic constraints and improve early-epoch model convergence.

PROJECTS

Resume Generator v1 | *Python, Jinja2, Textual, LaTeX* [github](#) 

- Built a modular pipeline to transform YAML resume data into Jinja2-rendered LaTeX artifacts.
- Added a Textual dashboard for config editing and generation without touching the pipeline internals.

Marketplace Platform | *NestJS, PostgreSQL, AWS*

- Led architecture and implementation of a production marketplace product from blank repo to deployment.
- Designed backend workflows that balanced product speed with transactional correctness.

SKILLS

Programming Languages: `built-in method items of dict object at 0x111372080;`

Frameworks: `built-in method items of dict object at 0x1113720c0;`

Platforms: `built-in method items of dict object at 0x111372000;`

PROFESSIONAL CERTIFICATES

Stanford Online December 2025
XCS224N Natural Language Understanding [credential](#) 